JAN KWONG

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EDUCATION

University of California, San Diego **La Jolla, CA**
M.S. in Computer Science (Anticipated)
Sep 2025 – Dec 2026 (GPA 3.8)
B.S. in Computer Science
Sep 2021 – Jun 2025 (GPA 3.8)

PROJECTS

picoLM: Compact Language Model **La Jolla, CA** **Apr 2026 – Jun 2026**
- Pretrained a 98M-parameter decoder-only transformer (GQA, SwiGLU, RoPE) in PyTorch on a single shared, interruptible GPU, achieving ~41 perplexity on a 5M-token web-text validation set
- Built a streaming data pipeline over FineWeb, SlimPajama, DCLM, and RedPajama-V2 with on-the-fly hash deduplication and token packing, never materializing the corpus on disk
- Applied knowledge distillation from GPT-Neo-2.7B, combining teacher-imitation and ground-truth losses

RAG & Context Engineering Pipeline **La Jolla, CA** **Apr 2026 – May 2026**
- Built a single-turn Q&A Retrieval-Augmented Generation system over a technical OSS documentation corpus, enforcing a strict 2,000-token per-query context budget
- Conducted a systematic sweep of 8+ configurations (varying chunk size, embedding model, retrieval strategy, and reranker settings) to optimize retrieval and generation quality
- Achieved 0.93 Correctness and 1.00 Faithfulness by implementing parent-document retrieval with BM25/vector hybrid search, a BGE cross-encoder reranker, and a dual-citation few-shot prompting strategy

Security Analysis of Apple Intelligence **La Jolla, CA** **Oct 2025 – Dec 2025**
- Reverse-engineered macOS binaries using Ghidra to reconstruct the information flow between user input, Generative AI models, and local storage
- Identified a privacy vulnerability where the system persisted unencrypted sensitive personal information to local directories, bypassing the ephemeral-only privacy policy
- Developed a Python forensic script to simulate TCC (Transparency, Consent, and Control) inheritance attacks, successfully exfiltrating sensitive user data from sandboxed environments

EXPERIENCE

UC San Diego **La Jolla, CA** **Apr 2026 – Jun 2026**
Graduate Teaching Assistant
- Led weekly discussion sections of ~40 students for the class Intro to Research Methods, mentoring them through empirical study design, statistical validation, and analysis methodologies
- Designed and refined discussion slides covering key course concepts, hosted weekly office hours, and gave students concrete feedback on assessments, helping them grasp difficult ideas and improve their work

UC San Diego LASR Lab **La Jolla, CA** **Jan 2024 – Dec 2024**
Undergraduate Research Assistant
- Co-authored a paper (submitted to Acoustical Society of America & PsyArXiv) benchmarking OpenAI's Whisper against 75 human transcribers on 300 English sentences from 20 diverse accented speakers
- Developed Python scripts to automate the computation of word error rates (WER), identifying that humans significantly outperform models on isolated words, suggesting potential limitations in models' training data or acoustic context requirements

SKILLS

- **Programming Languages:** Python, SQL, Java, C, C++, JavaScript, HTML, CSS, ARM, Clojure, Rocq
- **Libraries & Tools:** Pandas, NumPy, PyTorch, Scikit-learn, Git, Hugging Face, Kubernetes, Docker, Ghidra
- **Concepts:** Data Structures, Algorithms, Agile, CI/CD, Machine Learning, NLP, Computer Vision, Data Engineering, Data Analytics, Database Systems, Operating Systems, Computer Architecture, Computer Security
- **Languages:** Native/bilingual proficiency in Cantonese, English, and Mandarin; intermediate in Japanese